

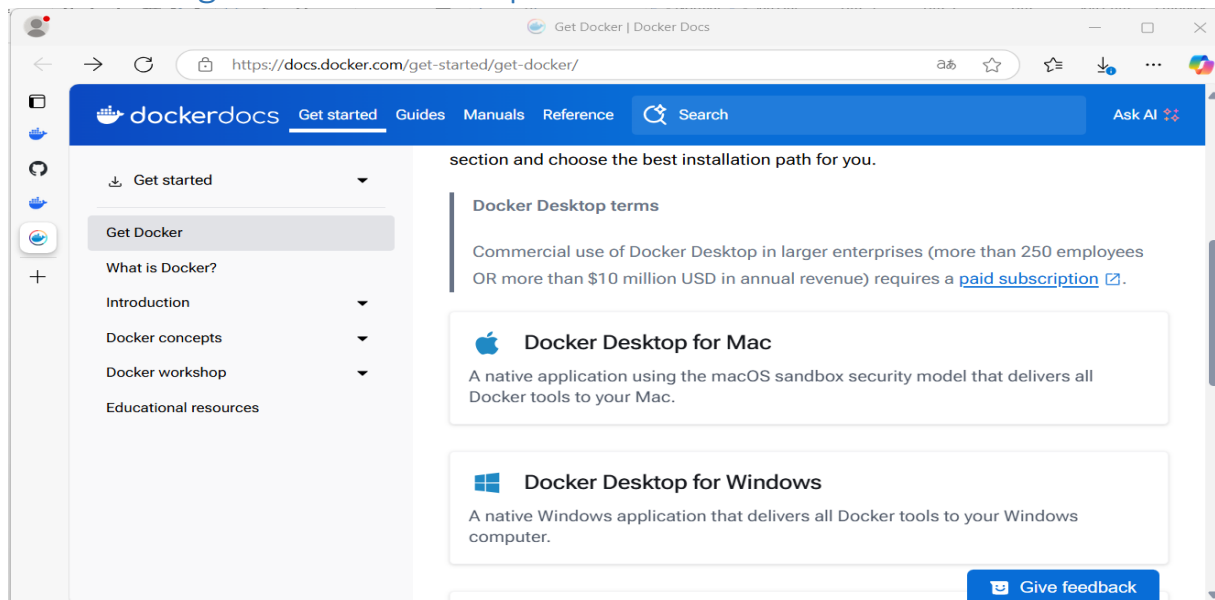
Atelier N° 9 : Master DSBD & IA

Pr. F. BENABBOU, Pr. A. DAIF

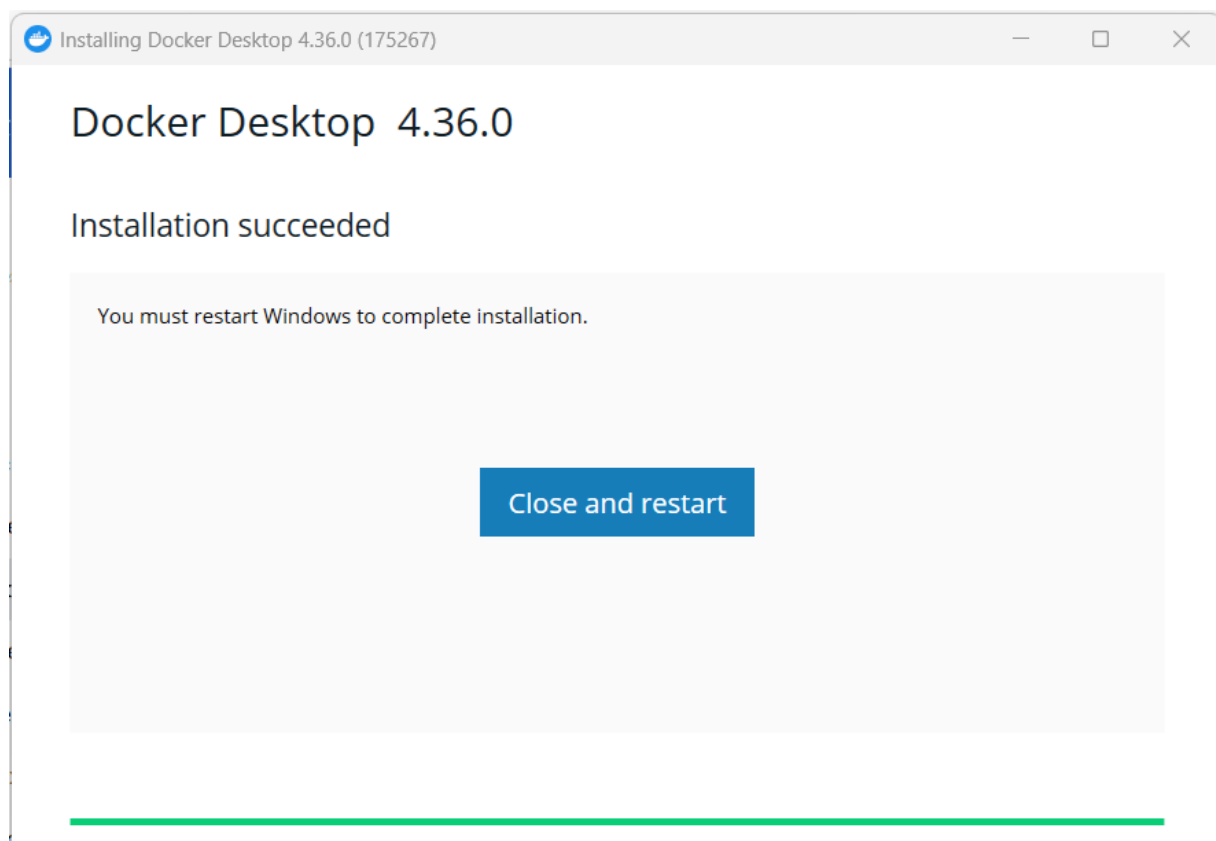
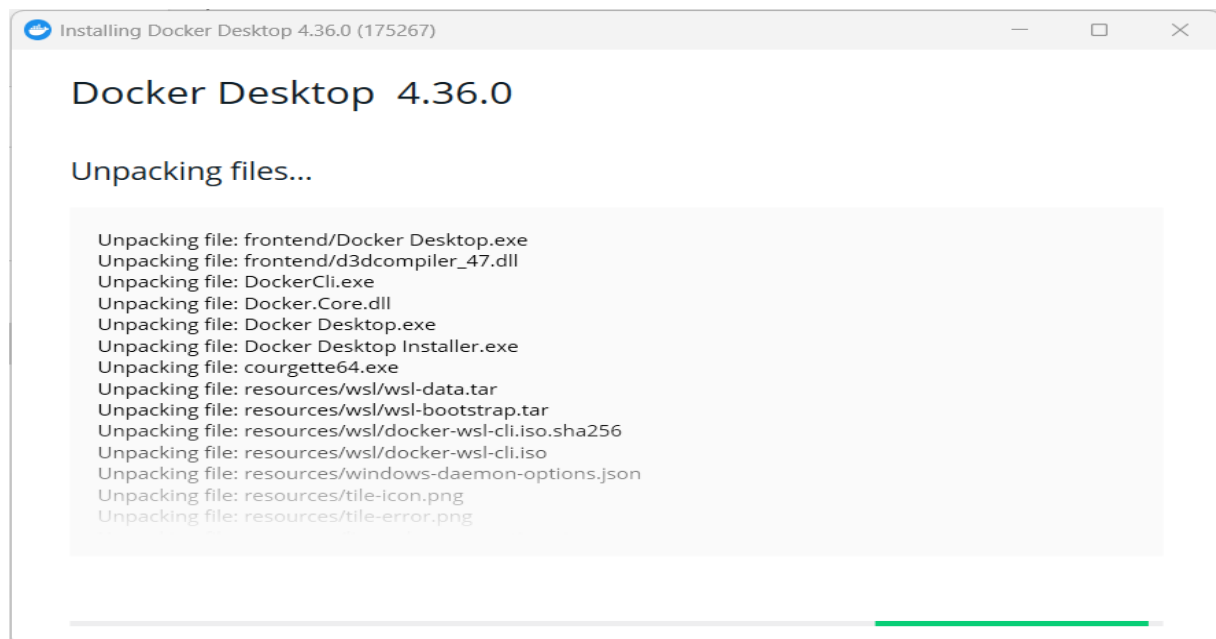
L'atelier N°9 a pour but de se familiariser Docker et la virtualisation des applications.

- L'installation Docker
- Conteneurisation d'une application et son déploiement.

1. Télécharger Docker desktop



1. Installer Docker:



2. Vérifier que wsl (Windows Subsystem for Linux) est bien installé

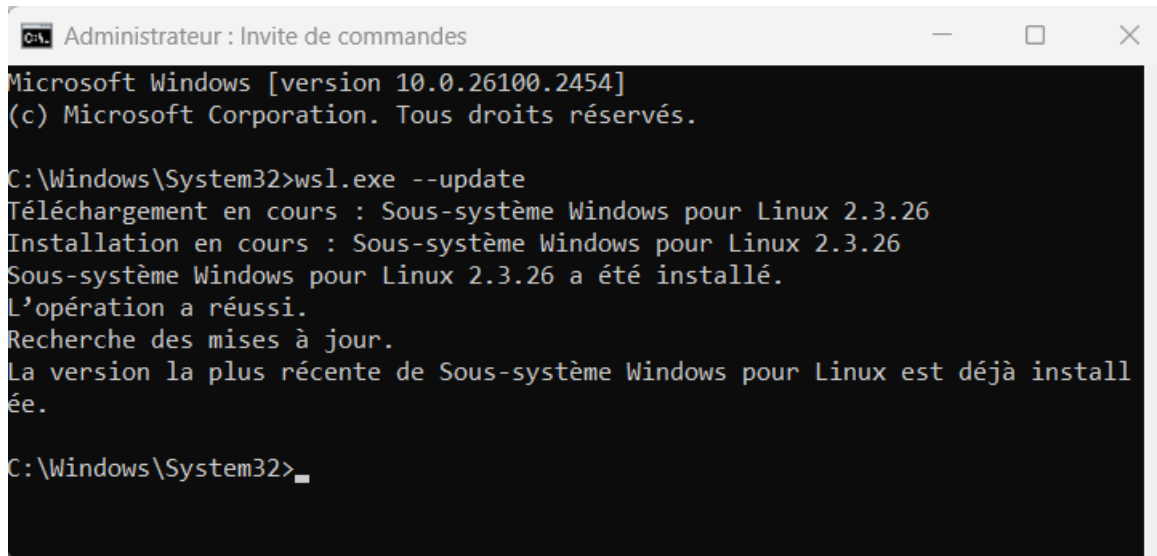
Il permet de faire tourner Linux directement dans Windows, sans machine virtuelle classique, sans dual-boot.



```
C:\Users\PC>wsl --status
Distribution par défaut : docker-desktop
Version par défaut : 2
C:\Users\PC>
```

3. Installer WSL.exe (sous système Windows pour linux)

Sinon Installez le ou faites un update..

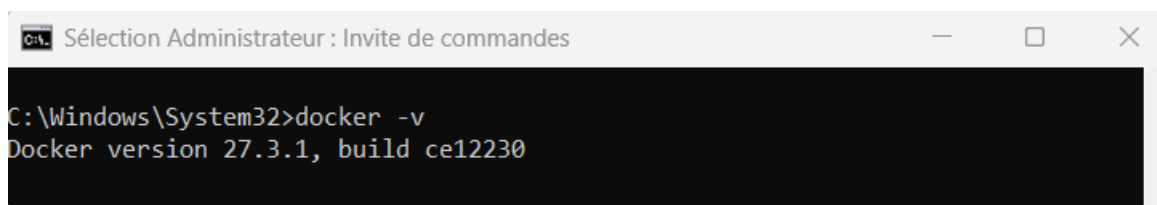


```
Microsoft Windows [version 10.0.26100.2454]
(c) Microsoft Corporation. Tous droits réservés.

C:\Windows\System32>wsl.exe --update
Téléchargement en cours : Sous-système Windows pour Linux 2.3.26
Installation en cours : Sous-système Windows pour Linux 2.3.26
Sous-système Windows pour Linux 2.3.26 a été installé.
L'opération a réussi.
Recherche des mises à jour.
La version la plus récente de Sous-système Windows pour Linux est déjà installée.

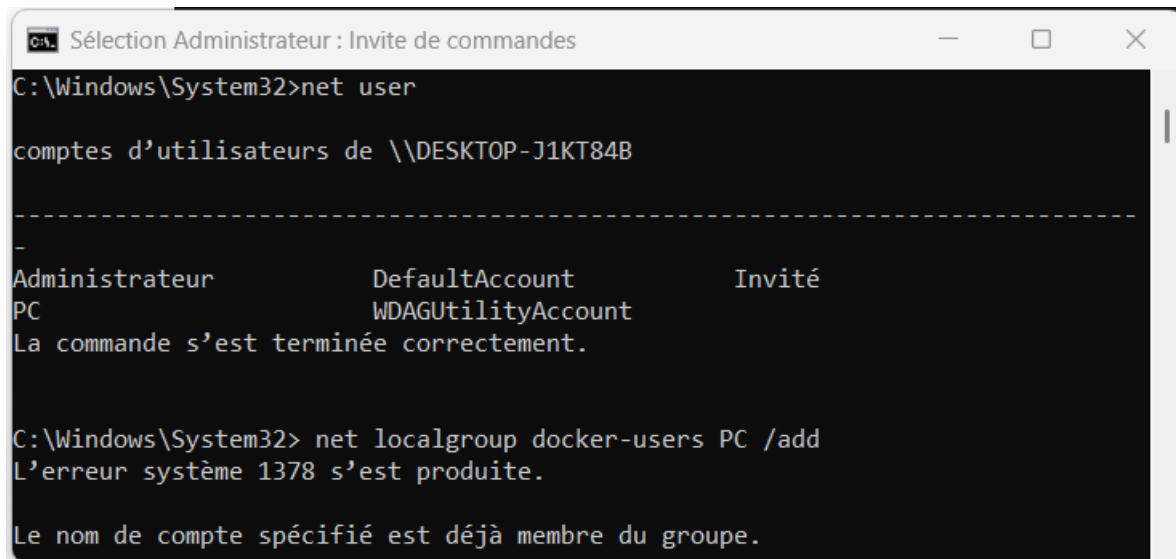
C:\Windows\System32>
```

4. Vérification de la version :



```
C:\Windows\System32>docker -v
Docker version 27.3.1, build ce12230
```

5. Verification des utilisateurs :



```
C:\Windows\System32>net user

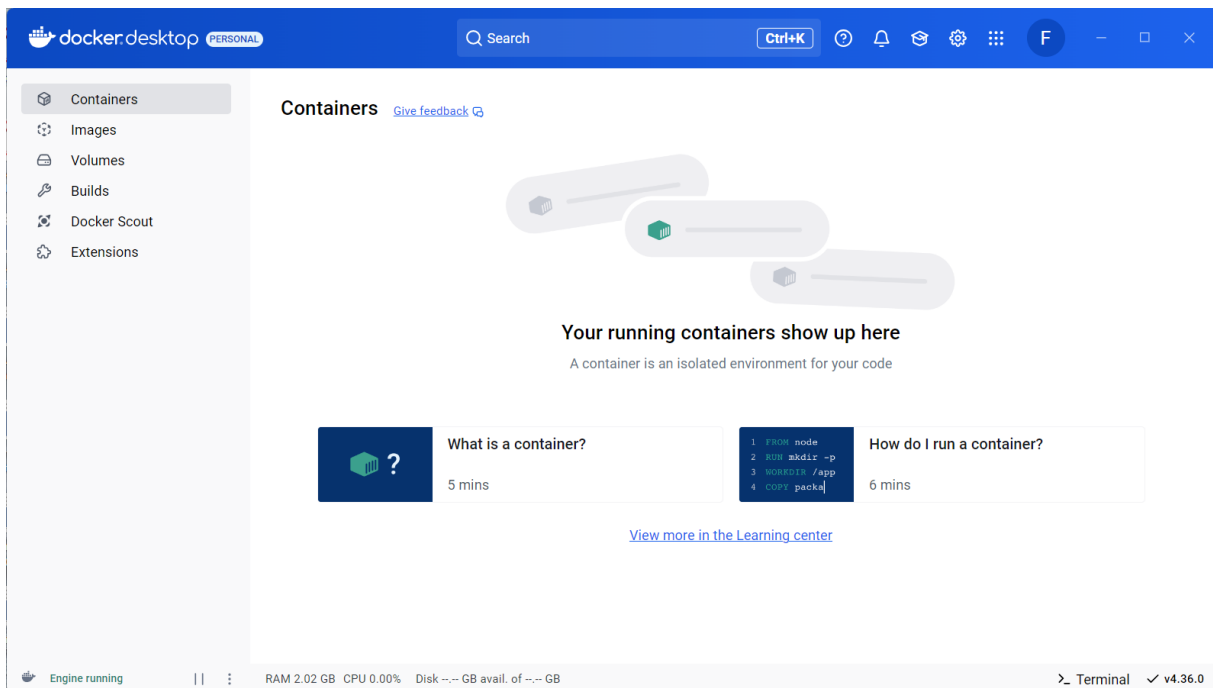
comptes d'utilisateurs de \\DESKTOP-J1KT84B

-----
-
Administrateur      DefaultAccount      Invité
PC                  WDAGUtilityAccount
La commande s'est terminée correctement.

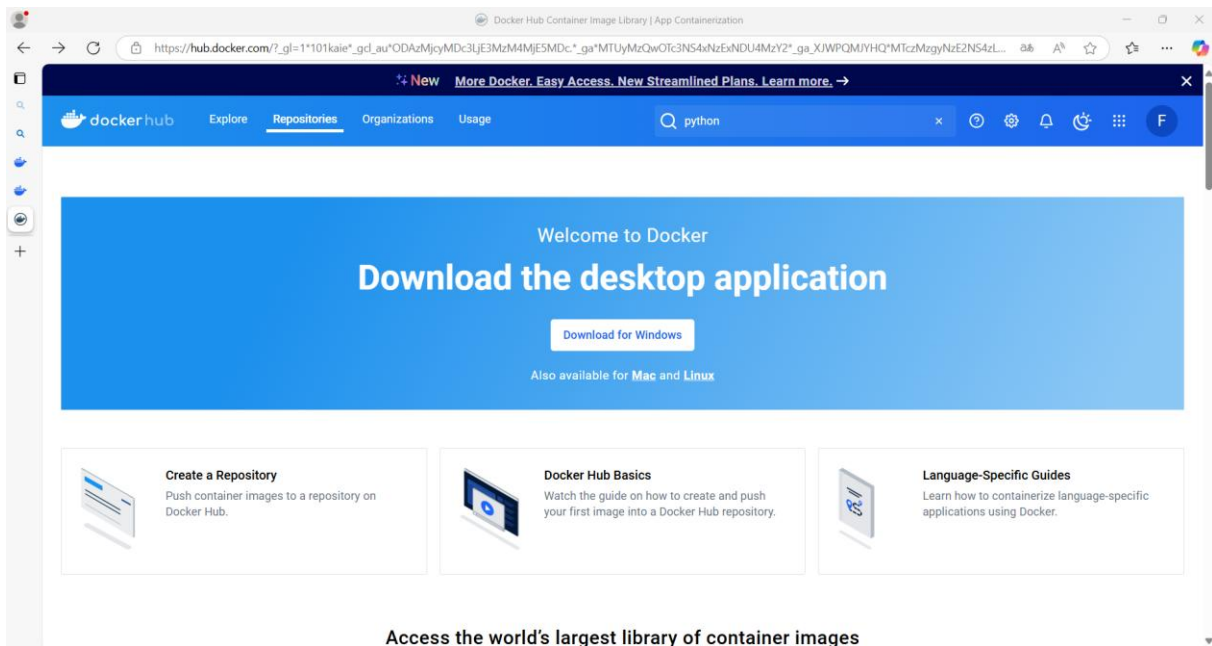
C:\Windows\System32> net localgroup docker-users PC /add
L'erreur système 1378 s'est produite.

Le nom de compte spécifié est déjà membre du groupe.
```

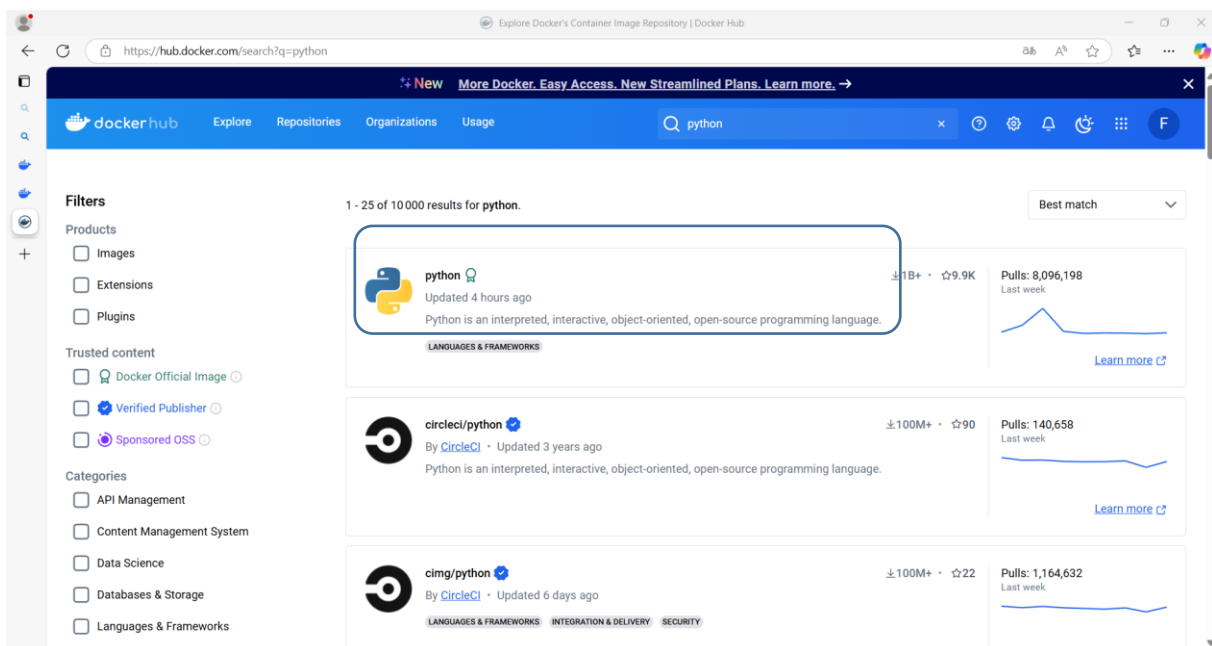
6. Lancement du Docker (raccourci sur bureau)



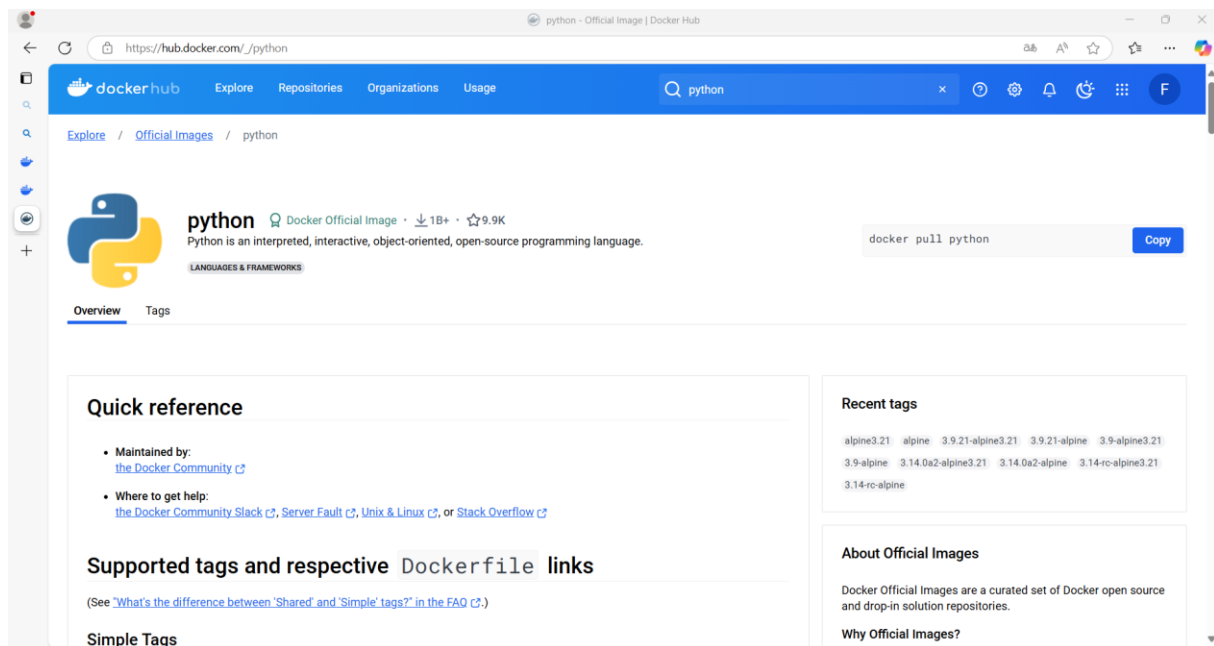
7. Docker hub



8. Chercher une image python :



9. Télécharger une image



Télécharger une image (Python) :

```
C:\windows\System32\Docker pull python
```

```
Administrateur : Invite de commandes

C:\Windows\System32> net localgroup docker-users Administrateur /add
La commande s'est terminée correctement.

C:\Windows\System32> docker start
"docker start" requires at least 1 argument.
See 'docker start --help'.

Usage:  docker start [OPTIONS] CONTAINER [CONTAINER...]

Start one or more stopped containers

C:\Windows\System32> docker pull python
Using default tag: latest
latest: Pulling from library/python
551df7f94f9c: Download complete
5bd71677db44: Download complete
ce82e98d553d: Download complete
2ac2596c631f: Download complete
abab87fa45d0: Download complete
5f0e19c475d6: Download complete
fdf894e782a2: Download complete
Digest: sha256:220d07595f288567bbf07883576f6591dad77d824dce74f0c73850e129fa1f46
Status: Downloaded newer image for python:latest
docker.io/library/python:latest

C:\Windows\System32>
```

On peut spécifier la version en utilisant le tag par défaut c'est « latest »

The screenshot shows the Docker Hub interface for the 'python' image. The 'Tags' tab is selected, displaying a list of available tags. The table includes columns for TAG, Digest, OS/ARCH, Vulnerabilities, and Compressed size. The 'alpine3.21' tag is highlighted as the latest pushed version.

TAG	Digest	OS/ARCH	Vulnerabilities	Compressed size
alpine3.21				
Last pushed 4 hours ago by dolanry				
	3a98017e1188	linux/386	None found	15.89 MB
	8f2ec77420c7	linux/amd64	None found	15.83 MB
	48799fcd5eb	linux/arm/v6	None found	15.17 MB
+4 more...				

10. Affichage des images :

C:\windows\System32\docker images

```
Administrateur : Invite de commandes

C:\Windows\System32>docker images
REPOSITORY    TAG       IMAGE ID      CREATED       SIZE
python        latest    220d07595f28  6 days ago   1.47GB

C:\Windows\System32>
```

11. Chercher une image :

C:\windows\System32\Docker search python

```
Administrateur : Invite de commandes

C:\Windows\System32>Docker search python
NAME                DESCRIPTION                                STARS   OFFICIAL
python              Python is an interpreted, interactive, objec... 9923    [OK]
circleci/python     Python is an interpreted, interactive, objec... 90
cimg/python         22
bitnami/python      Bitnami container image for Python           29
okteto/python       0
ubuntu/python       A chiselled Ubuntu rock with the Python runt... 18
paketobuildpacks/python 0
intel/python        0
chainguard/python   Build, ship and run secure software with Cha... 4
opensuse/python     openSUSE base image with python              0
corpusops/python    https://github.com/corpusops/docker-images/    1
stages/python       High-level scripting language                 0
metabrainz/python   0
jitesoft/python     Python & Pip on alpine.                       0
vulhub/python       1
jumpserver/python   Python is an interpreted, interactive, objec... 0
clarinpl/python     0
tensorchord/python  Base image for envd                           0
koxudaxi/python     Dockerfiles for CPython of lysnikolaou's tag... 0
dtcenter/python     Necessary Python3 packages for plotting GRIB... 1
openquantumsafe/python  Quantum-safe/liboqs crypto with python API    0
openeuler/python    0
nanozoo/python      0
python/bpo-patroni  0
python/docker-bpo   Docker image for local development of http://... 0
```

12. Supprimer une image :

C:\windows\System32\Docker rmi python

```
Administrateur : Invite de commandes

C:\Windows\System32> Docker rmi python
Untagged: python:latest
Deleted: sha256:220d07595f288567bbf07883576f6591dad77d824dce74f0c73850e129fa1f46

C:\Windows\System32> docker images
REPOSITORY    TAG       IMAGE ID      CREATED       SIZE

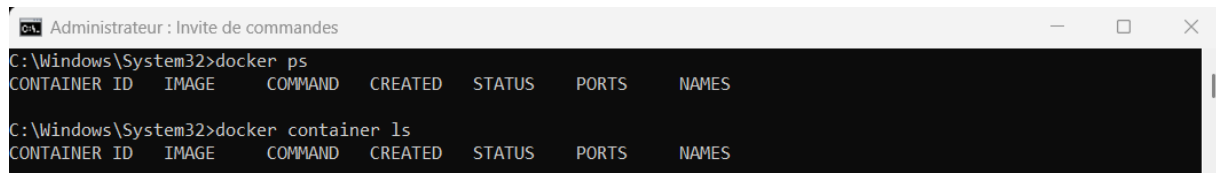
C:\Windows\System32>
```

13. Afficher les containers actifs (ici pas de conteneur)

Pour le moment il n'y a pas de conteneurs.

```
C:\windows\System32\Docker ps
```

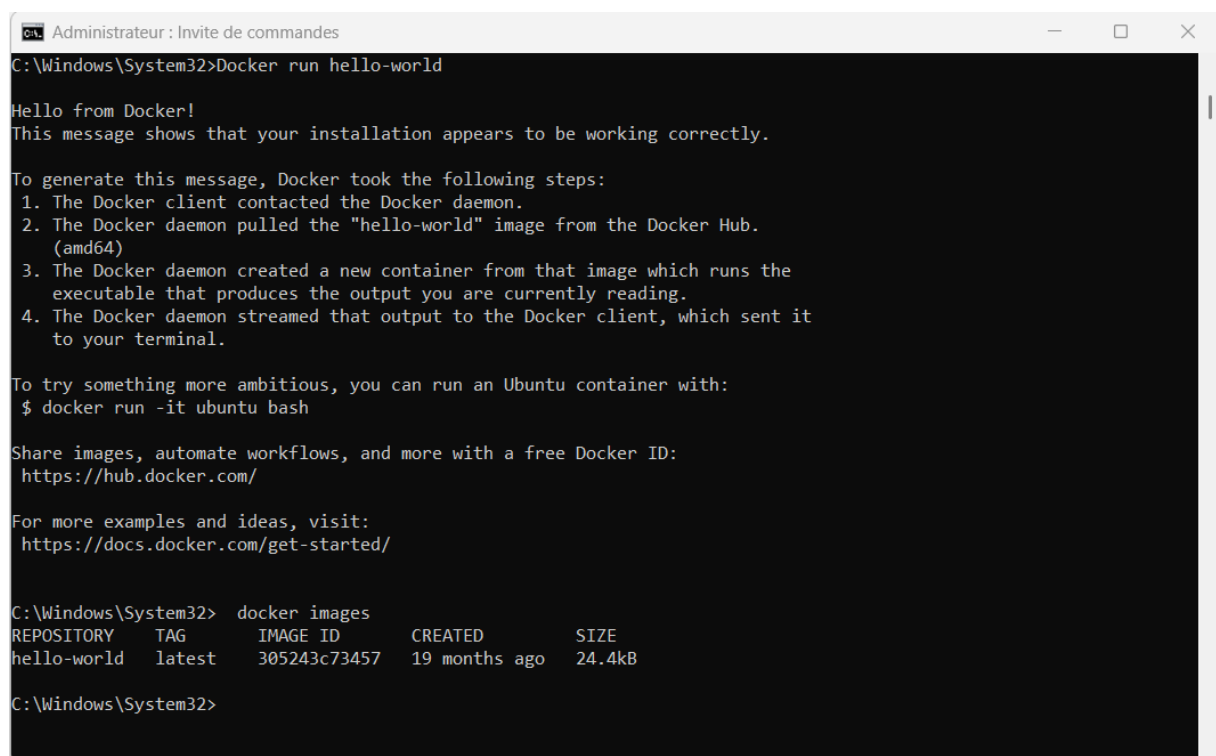
```
C:\windows\System32\Docker container ls
```



```
Administrateur : Invite de commandes
C:\Windows\System32>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
C:\Windows\System32>docker container ls
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
```

14. First app hello-world

Premier test pour vérifier que votre environnement est bien installé.



```
Administrateur : Invite de commandes
C:\Windows\System32>Docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

C:\Windows\System32> docker images
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE
hello-world    latest    305243c73457  19 months ago  24.4kB

C:\Windows\System32>
```

15. Déploiement d'une application node JS:

Dans mon cas l'application se trouve dans C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app. Ce répertoire contient :

```
Administrateur : Invite de commandes

Répertoire de C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app

10/12/2024 13:40 <DIR>      .
10/12/2024 13:40 <DIR>      ..
27/10/2020 16:53          22 .gitignore
27/10/2020 16:53      1 239 app.js
10/12/2024 13:40 <DIR>      backend
27/10/2020 16:53      127 Dockerfile
10/12/2024 13:40 <DIR>      fonts
27/10/2020 16:53      1 826 index.html
27/10/2020 16:53      1 131 LICENSE
27/10/2020 16:53      572 package.json
27/10/2020 16:53      888 readme.md
27/10/2020 16:53      1 071 server.js
27/10/2020 16:53      1 227 site.css
          9 fichier(s)          8 103 octets
          4 Rép(s) 118 219 837 440 octets libres

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

Le fichier Dockerfile :

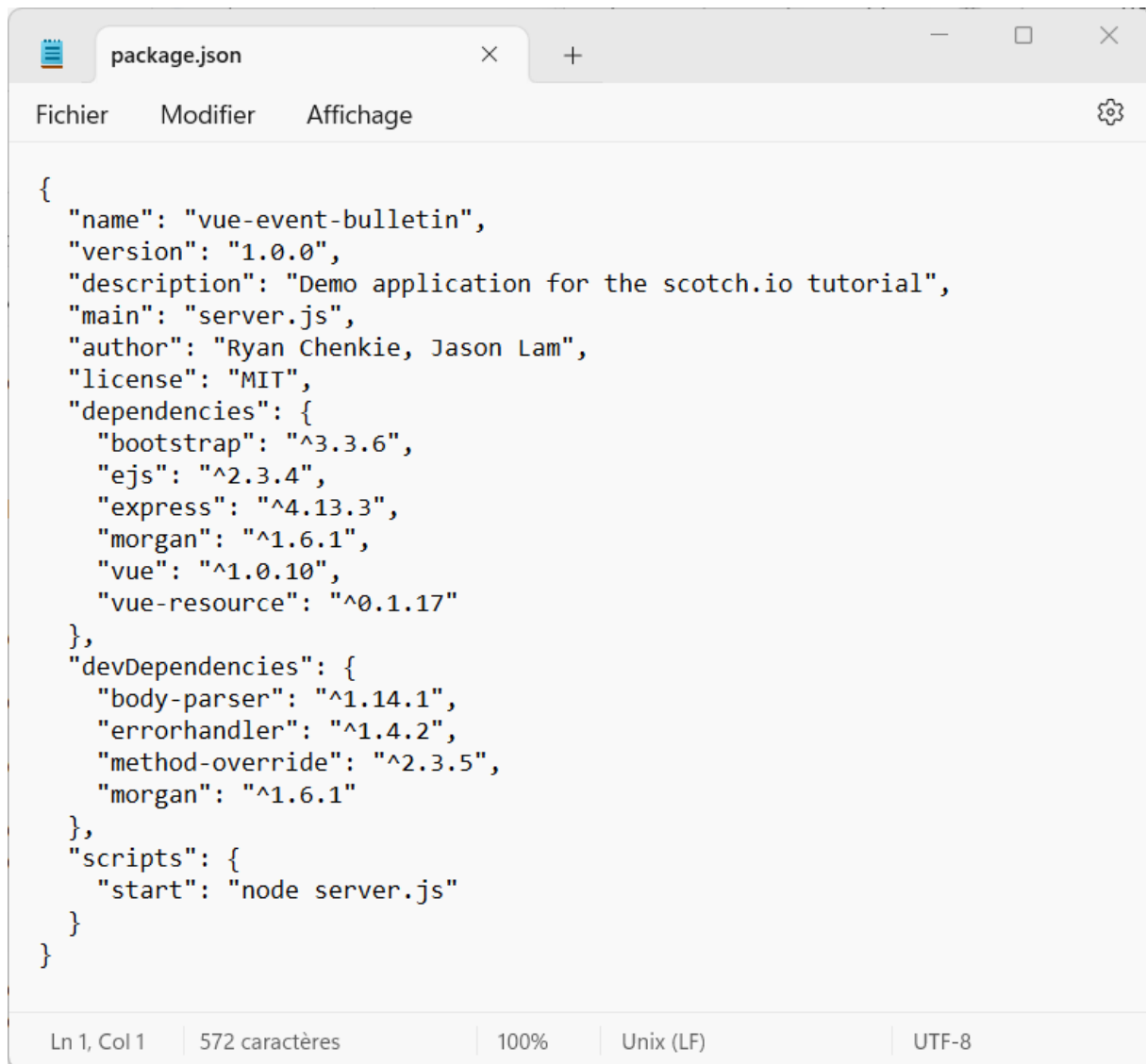
```
FROM node:current-slim

WORKDIR /usr/src/app
COPY package.json .
RUN npm install

EXPOSE 8080
CMD [ "npm", "start" ]

COPY . .
```

Le fichier json contient le **start** et les **dependances** :

A screenshot of a code editor window titled 'package.json'. The editor has a menu bar with 'Fichier', 'Modifier', and 'Affichage', and a settings icon on the right. The main area contains the JSON content of the package.json file. At the bottom, a status bar shows 'Ln 1, Col 1', '572 caractères', '100%', 'Unix (LF)', and 'UTF-8'.

```
{
  "name": "vue-event-bulletin",
  "version": "1.0.0",
  "description": "Demo application for the scotch.io tutorial",
  "main": "server.js",
  "author": "Ryan Chenkie, Jason Lam",
  "license": "MIT",
  "dependencies": {
    "bootstrap": "^3.3.6",
    "ejs": "^2.3.4",
    "express": "^4.13.3",
    "morgan": "^1.6.1",
    "vue": "^1.0.10",
    "vue-resource": "^0.1.17"
  },
  "devDependencies": {
    "body-parser": "^1.14.1",
    "errorhandler": "^1.4.2",
    "method-override": "^2.3.5",
    "morgan": "^1.6.1"
  },
  "scripts": {
    "start": "node server.js"
  }
}
```

16. Construction de l'image (« . » C'est pour dire que le fichier est local) :

```
C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app\docker build -t firstapp:1.0 -f Dockerfile .
```

Vous pouvez ajouter un tag (1.0)

```
Administrateur : Invite de commandes

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>docker build -t firstapp -f Dockerfile .
[+] Building 22.2s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 164B                                0.0s
=> [internal] load metadata for docker.io/library/node:current-slim 2.7s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:current-slim@sha256:8b30809f66a6ea8896b9a5d004b4fe2cc0e8061d981d3784fb 9.6s
=> => resolve docker.io/library/node:current-slim@sha256:8b30809f66a6ea8896b9a5d004b4fe2cc0e8061d981d3784fb 0.0s
=> => sha256:e486c8868d474e29e178b268b95b679ba2abb88eab17f1e04b0912e48e3a3103 448B / 448B 0.2s
=> => sha256:8030ee9caa1cdc982f1a2d504c88cc8f2d6b510f5b6b911e80ecc31b80eafa55 1.71MB / 1.71MB 0.4s
=> => sha256:35a89838774a5d20bde7b044149eaf3820e081656110dec3261a8504cbf61aa0 49.36MB / 49.36MB 7.1s
=> => sha256:8663e79f65016221dc2c9981c65291318e087053b0bbfdfd122d00ba573db12e 3.31kB / 3.31kB 0.8s
=> => sha256:bc0965b23a04fe7f2d9fb20f597008fcf89891de1c705ffc1c80483a1f098e4f 28.23MB / 28.23MB 6.6s
=> => extracting sha256:bc0965b23a04fe7f2d9fb20f597008fcf89891de1c705ffc1c80483a1f098e4f 1.1s
=> => extracting sha256:8663e79f65016221dc2c9981c65291318e087053b0bbfdfd122d00ba573db12e 0.0s
=> => extracting sha256:35a89838774a5d20bde7b044149eaf3820e081656110dec3261a8504cbf61aa0 1.4s
=> => extracting sha256:8030ee9caa1cdc982f1a2d504c88cc8f2d6b510f5b6b911e80ecc31b80eafa55 0.1s
=> => extracting sha256:e486c8868d474e29e178b268b95b679ba2abb88eab17f1e04b0912e48e3a3103 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 32.84kB                                  0.0s
=> [2/5] WORKDIR /usr/src/app                                    0.5s
=> [3/5] COPY package.json .                                    0.1s
=> [4/5] RUN npm install                                         6.6s
=> [5/5] COPY . .                                                0.1s
=> exporting to image                                           2.2s
=> => exporting layers                                           1.3s
=> => exporting manifest sha256:5b9472ebaef43223e4e57d3caaab7a98ccf922548e5cb01cd6ae534ae2de6f2c 0.0s
=> => exporting config sha256:e042ffe3b5acafa62214fefeceb6ef77c768089298d475e4e29096265e3fcf91 0.0s
=> => exporting attestation manifest sha256:0b62faf3c3571306dcfa9e49fce56cf24a50d4960c8dc62fa2c20fed63f2a9 0.0s
=> => exporting manifest list sha256:a9f050b924e6b312b6968f512475ac95fb509125909ed6cedb5447fb0eacc917 0.0s
=> => naming to docker.io/library/firstapp:latest                0.0s
=> => unpacking to docker.io/library/firstapp:latest             0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/z492sekb7dvaqwsa2ar4u6bo
```

17. Vérification de la création de l'image:

```
Administrateur : Invite de commandes

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
firstapp      latest   a9f050b924e6   8 minutes ago  369MB
hello-world   latest   305243c73457   19 months ago  24.4kB

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

18. Création et démarrage d'un conteneur à partir de l'image firstapp

```
docker run -d -p 8180:8080 firstapp
```

```
Administrateur : Invite de commandes

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>docker run -d -p 8180:8080 firstapp
ec4419f8d75e084c30c9f7a795f185c682525a20e3064f8d3ebd548eeebfddbe

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

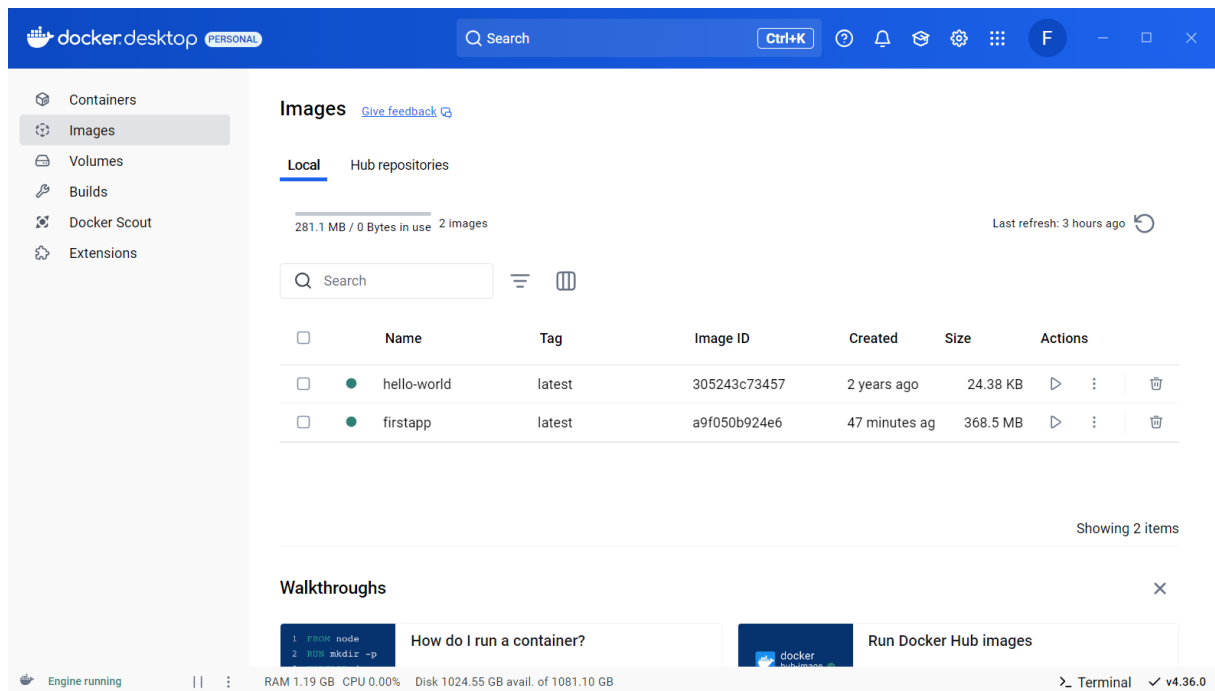
19. Vérification de création du container :

```
Administrator: Invite de commandes
C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
ec4419f8d75e   firstapp   "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  0.0.0.0:8180->8080/tcp    modest_n
ash

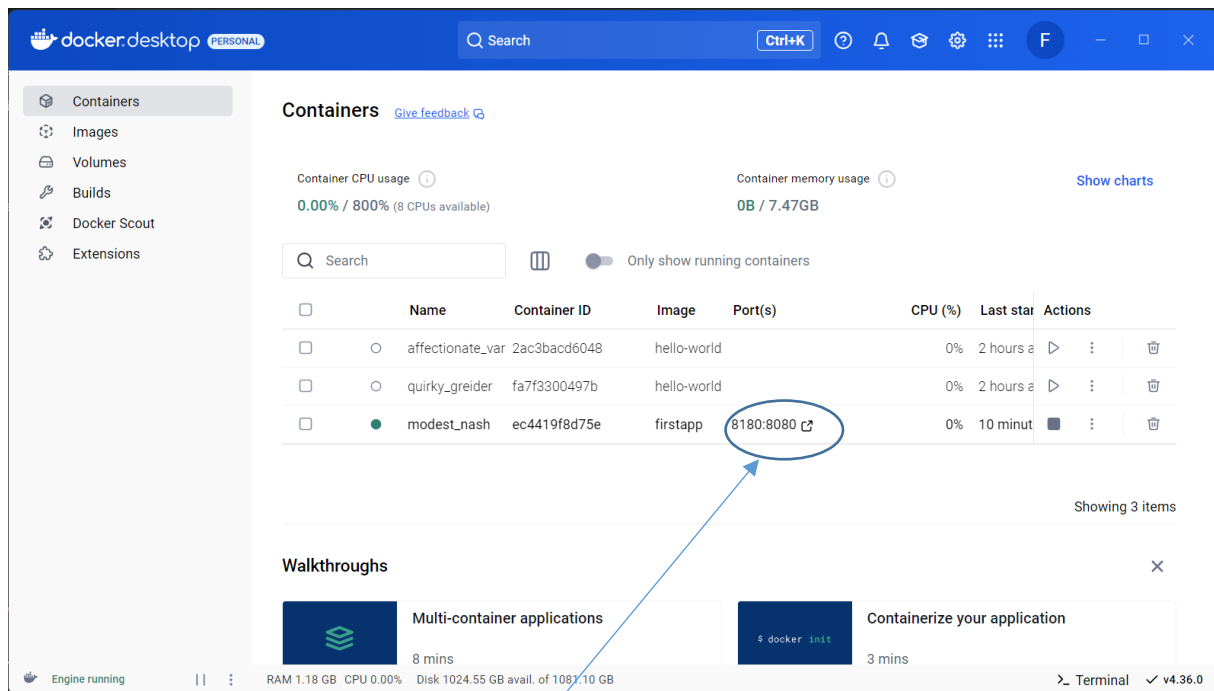
C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

20. Utiliser l'interface Docker de gestion :

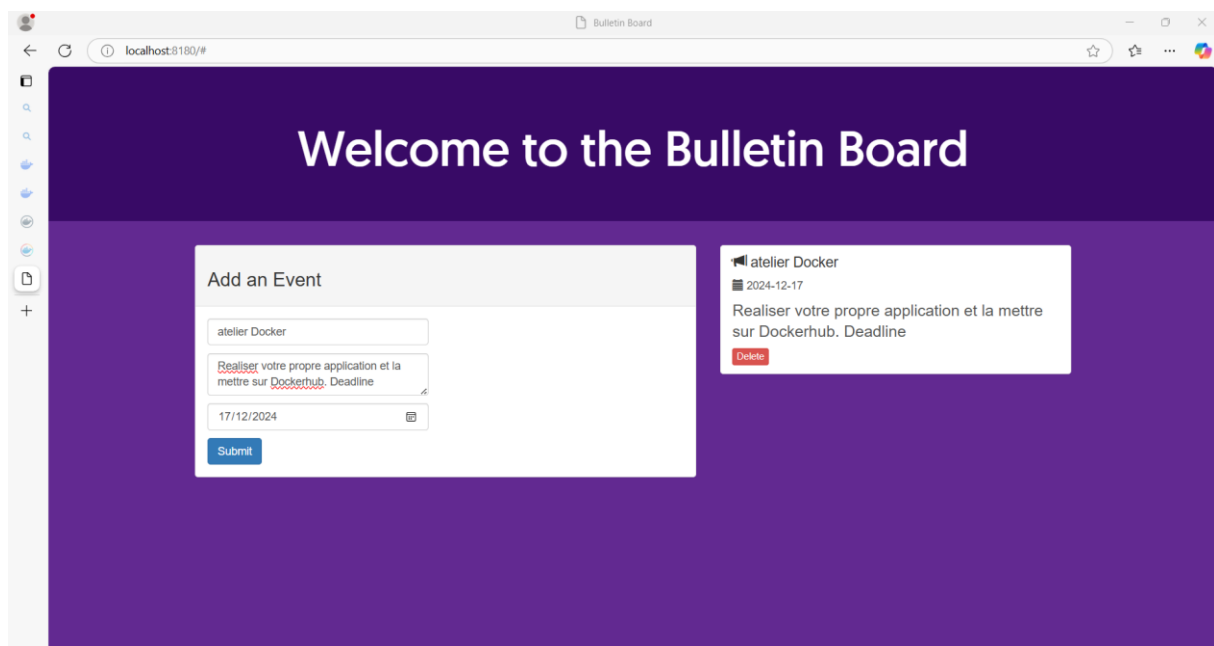
Lister les images :



Lister les containers :



21. Accès à l'application :



22. Push l'image dans le docker hub repository:

```
docker tag firstapp faouziabenabbou/firstapp
docker push faouziabenabbou/firstapp
```

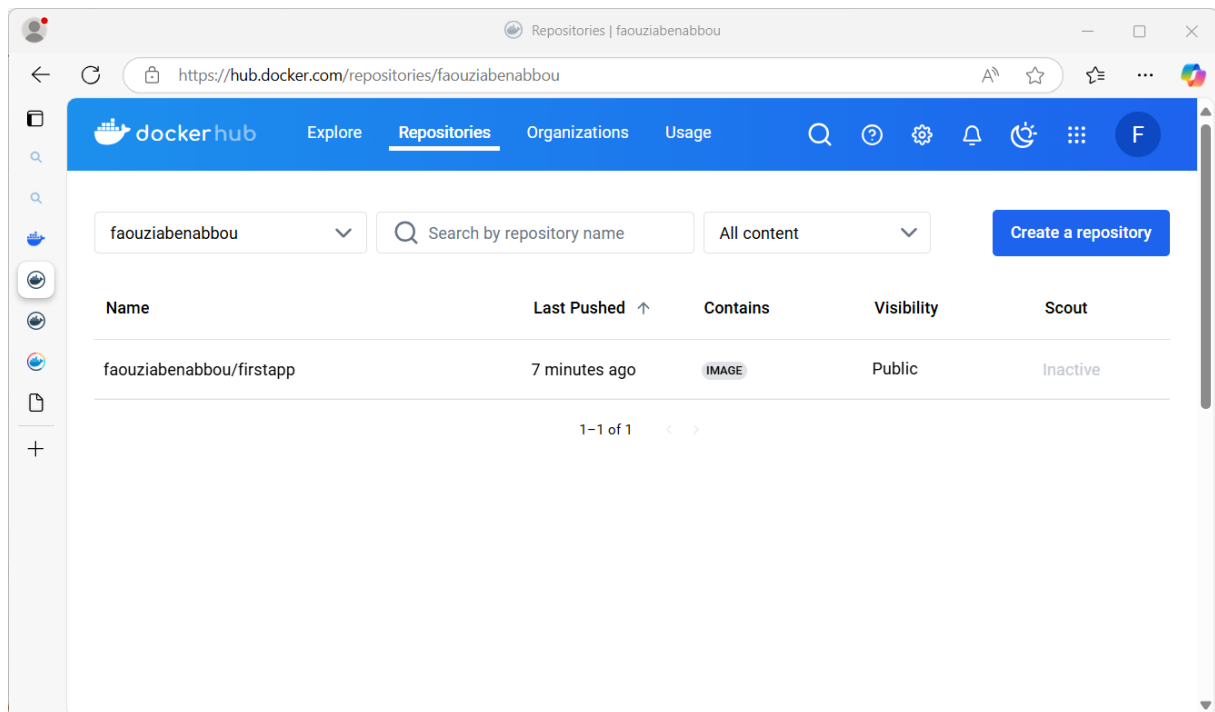
```
Administrateur : Invite de commandes

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker tag firstapp faouziabenabbou/firstapp

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app> docker push faouziabenabbou/firstapp
Using default tag: latest
The push refers to repository [docker.io/faouziabenabbou/firstapp]
b93e152f9b69: Pushed
3bf5736c596c: Pushed
6e80ef341530: Pushed
35a89838774a: Pushed
5d537f5db42c: Pushed
3e2b426e5e71: Pushed
8030ee9caa1c: Pushed
e486c8868d47: Pushed
8663e79f6501: Pushed
bc0965b23a04: Pushed
latest: digest: sha256:a9f050b924e6b312b6968f512475ac95fb509125909ed6cedb5447fb0eacc917 size: 856

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

Sur Docker hub:



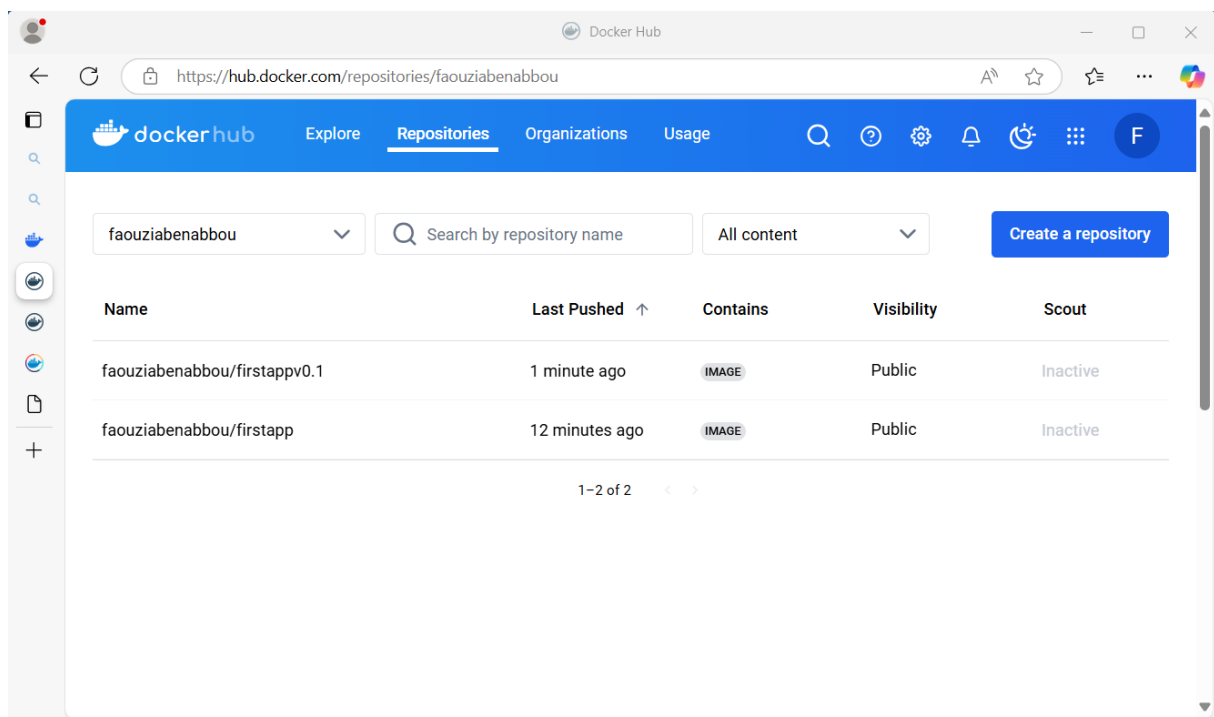
Une autre instance:

```
Administrateur : Invite de commandes

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>docker push faouziabenabbou/firstappv0.1
Using default tag: latest
The push refers to repository [docker.io/faouziabenabbou/firstappv0.1]
3e2b426e5e71: Mounted from faouziabenabbou/firstapp
35a89838774a: Mounted from faouziabenabbou/firstapp
3bf5736c596c: Mounted from faouziabenabbou/firstapp
3663e79f6501: Mounted from faouziabenabbou/firstapp
e486c8868d47: Mounted from faouziabenabbou/firstapp
5d537f5db42c: Mounted from faouziabenabbou/firstapp
8030ee9caa1c: Mounted from faouziabenabbou/firstapp
p93e152f9b69: Mounted from faouziabenabbou/firstapp
6e80ef341530: Mounted from faouziabenabbou/firstapp
bc0965b23a04: Mounted from faouziabenabbou/firstapp
latest: digest: sha256:a9f050b924e6b312b6968f512475ac95fb509125909ed6cedb5447fb0eacc917 size: 856

C:\Users\PC\Desktop\docker\node-bulletin-board-master\bulletin-board-app>
```

Sur docker hub:



L'image est publique sur mon compte, donc vous pouvez faire un pull et l'utiliser.

Résumé des étapes :

1- Téléchargement et installation de docker

<https://www.docker.com/products/docker-desktop/>

2- Commandes de base

Afficher l'aide de Docker

```
docker --help
```

Afficher la version Docker

```
docker --version
```

3. Gestion des images :

Docker Hub

<https://hub.docker.com/>

Téléchargement d'image (depuis le docker Hub)

```
docker pull Python
docker pull tomcat
docker pull tomcat:9.0.73-jre8
```

Partager une image local a docker hub

```
docker push < image >
```

Afficher la liste d'images disponibles

```
docker images
```

Recherche d'image

```
docker search <mot clé>
```

Créer une image docker à partir d'un Dockerfile

```
docker build
```

Supprimer une image

```
docker rmi < image >
```

4. Gestion des containers

Afficher les conteneurs actif

```
docker container ls
docker ps
```

Afficher les conteneurs en tâche de fond

```
docker ps -a
```

Créer un conteneur à partir d'une image

```
docker create <image>
```

```
docker run <image>
```

Démarrer un conteneur

```
docker start <conteneur>
```

Stopper un conteneur

```
docker stop <conteneur>
```

```
docker kill <conteneur>
```

Effacer un conteneur

```
docker rm <conteneur>
```

2- Application sur un projet node JS

- Savoir l'environnement nécessaire pour exécuter votre application
- Création de Dockerfile :
- Se déplacer vers le dossier de projet : `Cd < chemin de projet >`

Création d'une image à partir d'un dockerfile :

```
docker build -t firstapp -f Dockerfile .
```

Création et démarrage d'un conteneur à partir de l'image firstapp :

```
docker run -d -p 8180:8080 firstapp
```

Push l'image dans le docker hub:

```
docker tag firstapp votre_login/firstapp
```

```
docker push votre_login/firstapp
```